

Travel-Time Tomography: Inversion of First-Arrival Data to Retrieve Velocity-Depth Models

The report's structure and theoretical explanation are its main strengths . The synthetic test is a major weakness: the inverted model completely fails to recover the sharp, true layered model, a failure the report never discusses. The predicted travel times in the synthetic test are a poor fit, smoothing over the sharp crossover points of the input data. The Koenigsee result (Fig 6) is a clear U-shaped basin, but the text oversimplifies this as a three-layered structure. Table 1 incorrectly lists the units for slowness as seconds (s).

1. **The synthetic layered model is replaced by a crosshole synthetic test** using a uniform background model with embedded anomalies and the original layered model is now pushed to the appendix.

Appendix

The following section has been added to the appendix for the purposes of demonstrating modeling using PyGIMLi highlighting library usages. *We do however acknowledge that raytracing for a layered model has an analytical solution where care is taken to account for direct and critically refracted waves (governed by Snells law), which here is taken as an approximation of the raypath that is found by finding the shortest-path through a grid of nodes.*

2. **Justifications for the choice of reference model** for crosshole are now explicitly discussed in the report.

Rather than using the fine forward mesh for inversion, a coarser structured grid is constructed:

Table 4: Inversion mesh specifications

Parameter	Value	Description
X-extent	[0, 20] m	28 grid points
Y-extent	[0, -28] m	39 grid points
Cell spacing	0.75 m	Regular rectangular cells

Convergence stops when $\chi^2 \leq 1$. A homogeneous starting model is used `useGradient=False`. This results in:

```
invmodel = mgr.invert(data, mesh=mesh, secNodes=3, lam=1000,
                      zWeight=1.0, useGradient=False,
                      verbose=True)
```

$$v_0(\mathbf{r}) = 2000 \text{ m/s} \quad \forall \mathbf{r}$$
$$s_0(\mathbf{r}) = \frac{1}{2000} = 5.0 \times 10^{-4} \text{ s/m} \quad \forall \mathbf{r}$$

This choice of homogeneous starting model is justified for crosshole geometry: velocity typically does NOT vary systematically with depth between boreholes, so a gradient model (appropriate for surface surveys) is not used.

3. **Clarified Koenigsee Result Description Figure 3**

Final Inversion Map

Final results are available at travel-time-tomography/data/TravelTimeManager.

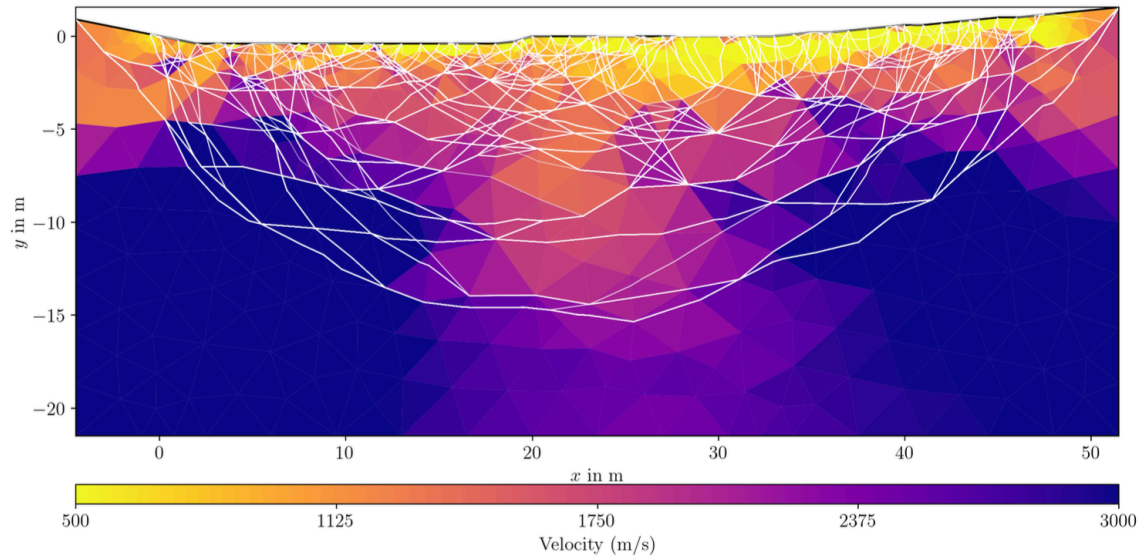


Figure 3: Final inverted velocity model for Koenigsee dataset. Color scale shows seismic P-wave velocity (m/s). The result reveals a broad U-shaped low-velocity zone flanked by higher-velocity regions, consistent with basinal infill above deeper bedrock, rather than a horizontally stratified or simply layered profile. Superimposed ray coverage (or ray paths) indicates region-specific illumination by the data.

4. The synthetic datasets - Crosshole and Layered are added for the purposes of demonstration while the main focus is still on the Koenigsee data inversion.
5. The table row with the slowness unit issue (s instead of s/m) has been removed.